

Project Title:

# Library Management System

Proposed By:

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# 1 Introduction

The Library Management System is a database project designed to organize and manage library operations efficiently.

The system helps store information about:

- Books
- Members
- Borrowing operations

It allows librarians to:

- Add books and members
- Track borrowed books
- Monitor return dates
- Manage available copies

The project is implemented using SQL and includes:

- ER Diagram
- Database Schema
- Tables with Relationships
- SQL Queries

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## Problem Statement

Managing library records manually can cause:

- Data loss
- Difficulty tracking borrowed books
- Errors in member records
- Time consumption

This system solves these problems by using a relational database.

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## Objectives

The main objectives of the project are:

- Store book information
- Store member information
- Record borrowing transactions
- Track return dates
- Maintain relationships between tables
- Improve data organization

# 4 Tools & Technologies

- Database Language: SQL
- DBMS: MySQL
- Design Diagrams: ERD & Schema Diagram

# 5 System Entities

## A) Books Entity

- Stores information about books.
- Attributes:
- Book\_ID (Primary Key)
- Title
- Author
- Category
- Available\_Copies

## B) Members Entity

Stores information about library members.

Attributes:

- Member\_ID (Primary Key)
- Name
- Email
- Phone

## C) Loan Entity

Stores borrowing operations.

Attributes:

- Loan\_ID (Primary Key)
- Book\_ID (Foreign Key)
- Member\_ID (Foreign Key)
- Borrow\_Date
- Return\_Date

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# Relationships

- Member — BORROWS — Loan
- One member can borrow many books.
- Relationship: One-to-Many
- Book — FOR — Loan
- One book can appear in many loan records.
- Relationship: One-to-Many

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## ER Diagram

The ER Diagram represents:

- Entities
- Attributes
- Relationships between tables

The project contains:

- Member Entity
- Book Entity
- Loan Entity
- BORROWS Relationship
- FOR Relationship

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## Database Schema

- The schema shows:
- Primary Keys (PK)
- Foreign Keys (FK)
- Relationships between tables
- Tables:
- Members
- Books
- Loan

# 9 SQL Implementation

The project includes:

- Creating Database
- Creating Tables
- Inserting Data
- Running Queries
- Join Operations

# 10 Advantages of the System

- Easy data management
- Accurate records
- Faster searching
- Organized relationships
- Reduces manual errors

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## Future Enhancements

Future improvements may include:

- Login system
- Admin dashboard
- Fine calculation for late returns
- Book reservation feature
- GUI application

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## Conclusion

- The Library Management System provides a simple and effective solution for managing library operations using SQL databases.
- It demonstrates the use of:
  - Relational databases
  - Primary & Foreign Keys
  - ER Diagrams
  - SQL Queries
  - Table Relationships
- The project improves organization and efficiency in managing library data.



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**Thank You**